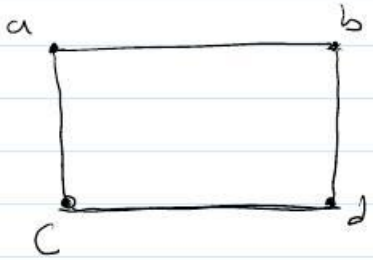
Tutorial 4: Graph Theory I

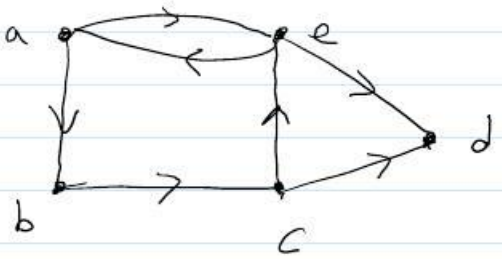
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Question 1: a) Simple graphs of undirected graphs have no multiple edges or loops. Thus,



← after removing multiple edges and loops

b) A simple undirected graph has no repeated directed edges or loops.



Question 2:

$$v = 9 \quad , \quad m = 12$$

$$\deg(a) = 3 \quad , \quad \deg(c) = 4 \quad , \quad \deg(e) = 6$$
$$\deg(b) = 2 \quad , \quad \deg(d) = 0 \quad , \quad \deg(g) = 4$$
$$\deg(h) = 2 \quad , \quad \deg(i) = 3 \quad , \quad \deg(f) =$$

isolated points $\{f, d\}$, $\deg(v) = 0$.

There are no vertices of degree 1 Thus, no pendant vertice

Question 3: since $2m = \sum \deg(v) = 5 \times 15$ ← not even

Thus, no such graph can be constructed.

Question 4: a) $v = 4$, $m = 8$

$\deg^-(a) = 2$, $\deg^+(a) = 2$
$\deg^-(b) = 3$, $\deg^+(b) = 4$
$\deg^-(c) = 2$, $\deg^+(c) = 1$
$\deg^-(d) = 1$, $\deg^+(d) = 1$

b) $v = 5$, $m = 13$

$\deg^-(a) = 6$ $\deg^+(a) = 1$
$\deg^-(b) = 1$ $\deg^+(b) = 5$
$\deg^-(c) = 2$ $\deg^+(c) = 5$
$\deg^-(d) = 4$ $\deg^+(d) = 2$
$\deg^-(e) = 0$ $\deg^+(e) = 0$

Question 5: let $v \geq 2$, since

$$2m = \sum \deg(v)$$

Thus, $0 \leq \deg(v) \leq n-1$ for any vertex
(not connected; ↑ maximum number of connections.)

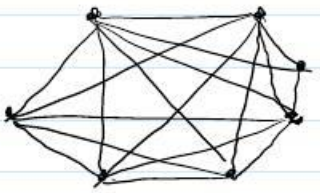
if $\exists v_1$ such that $\deg(v_1) = 0$, then $\nexists v_2$ such that $\deg(v_2) = n-1$ since v_1 is not adjacent to any vertex while v_2 is adjacent to every vertex $\Rightarrow$ a contradiction.

$\Rightarrow$ $0 \leq \deg(v) < n-1$ or $0 < \deg(v) \leq n-1$ $\forall v$.

notice that we have n vertices while $\deg(v)$ can only have $(n-1)$ different values $\Rightarrow$ at least two of them are the same.

Question 6:

a)



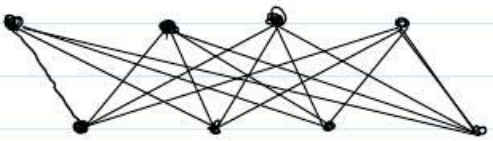
$n = 7$

$m = \frac{7(7-1)}{2} = 21$

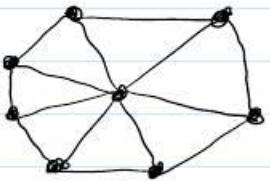
b) $n = 7$, $m = 7$



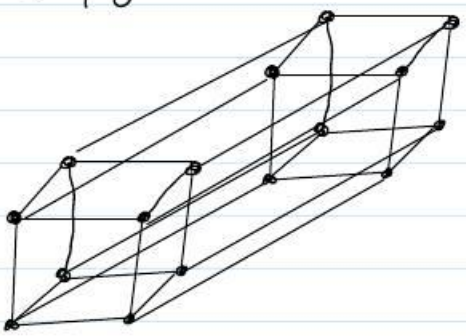
c)



d)

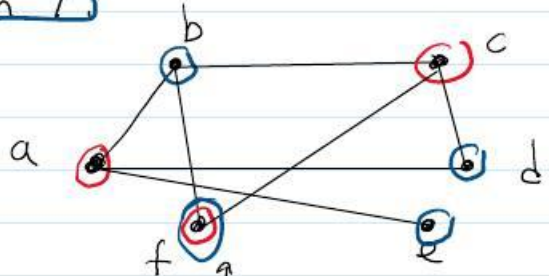


e) $r = 2^4 = 16$



Question 7:

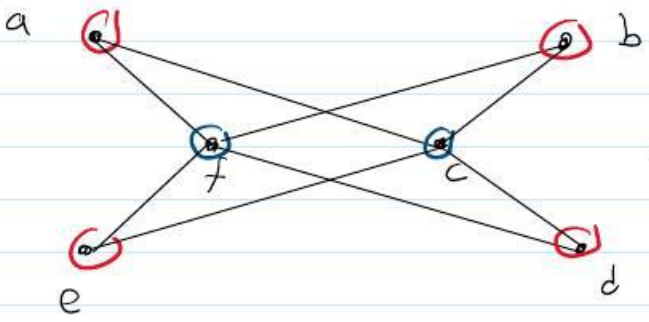
a)





contradiction ← not bipartite.

b)



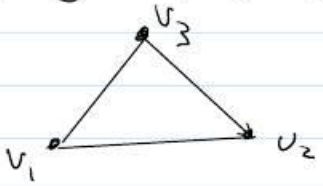
← is bipartite.

Question 8

a) $n=1$ K_1 is not bipartite $n \neq 2$

$n=2$ K_2 bipartite

$n>2$ since every edge is connected to the other edge. Thus, by default ∃ triangle graphs of the forms



as subgraphs

and since this subgraph cannot be bipartite ∀ $n>2$, there is no bipartite graphs.

b) C_n, $n \geq 3$ for C_n to be defined

let consider two examples



C_3

is not bipartite



C_4

is bipartite.

it seems that if n is odd C_n is not bipartite while if n is even C_n is bipartite

proof: suppose G is a cycle graph with n even. let

while if n is even C_n is bipartite

proof: suppose G is a cycle graph with n even. let v_0 be any vertex in the graph. Notice that if a path transcends 1 vertex color would change. Assign 1 to vertex v_0, each time it transcends a vertex multiply by (-1) to repeat color changes. Thus, moving to the next adjacent vertex we get $1(-1)$, thus color changes. Now, to move along the cycle, we transcend n vertices. thus, we get

$(-1)^n$ thus, if n is even, then it will change color making the graph bipartite. If not v_0 would have two colors which is a contradiction.

c) W_n since $\forall n \; K_3 \subseteq W_n \Rightarrow \forall n \; W_n$ is not bipartite

d) $Q_n \; \forall n \geq 1$ Q_n is bipartite this is clear by induction. Since Q_n is a connection between of 1-D between two Q_{n-1} graphs $\Rightarrow Q_n$ would be bipartite if correspondig vertices have different colors and each Q_{n-1} is bipartite. Since this is a recursive formula. we only need to show that Q_1 is bipartite and this is true.

Question 9: Let (V_2, m_2) be graph induced by set V_2 of vertices of complete graph K_n. by definition since every two vertices of K_n are connected $\Rightarrow$ any subset of V would be connected thus, (V_2, m_2) is complete.

Question 10:

a) vertices n

edges: Since, in K_n, every vertex is of degree $n-1 \Rightarrow$

$2 \cdot E: \sum \deg(v) = n(n-1) \Rightarrow E = n(n-1)/2$

b) vertices m+n

edges: since there are n vertex with degree m and m vertex

edges : since there are n vertices with degree m and m vertices with degree $n \Rightarrow 2E = mn + nm \Rightarrow E = mn$.

c) Vertices n
edges : since every vertex is of degree (2) $\Rightarrow E = n$.

d) vertices: by definition Q_n connects two Q_{n-1} graphs
thus, $V_n = 2V_{n-1} \Rightarrow V_n = 2^{n-1} V_1$
since $V_1 = 1 \Rightarrow V_n = 2^n$ vertices
edges ; by definition,

edges in Q_n = 2 edges in Q_{n-1} + edges of connection.
↑ just number of vertex of Q_{n-1}

$$\Rightarrow E_n = 2E_{n-1} + 2^{n-1}$$
$$= 2(2E_{n-2} + 2^{n-2}) + 2^{n-1}$$
$$= 4E_{n-2} + 2(2^{n-1})$$
$$\vdots$$
$$= 2^{n-1} E_1 + (n-1)2^{n-1}$$

since $E_1 = 1 \Rightarrow E_n = 2^{n-1} + (n-1)2^{n-1} = n2^{n-1}$

e) vertices : $n+1$
edges : edges of $C_n + 1 = n+1$